

Project 1
CSE4110 and AIE4055
Database Systems
Spring 2026

In this course project, you will design a **conceptual schema** for an **automobile company** enterprise using the **Entity-Relationship (E-R) model**. No implementation work (e.g., relational schema creation, coding, or database population) is required. This project focuses **exclusively** on developing a comprehensive E-R diagram.

Submission Rules

- Late submissions will not be accepted.
- Due date: Tuesday, April 14, 2026, 23:59.
- Submit both a report (PDF) and the E-R diagram (PNG).
- Upload your submission to the assignment section on Cybercampus.

Goal

The goal of this project is to provide a realistic experience in the *conceptual design* of a relational database for an automobile company. You will work **individually** to create an E-R diagram that accurately models the operations of an automobile company. This includes modeling vehicles, brands, models, suppliers, customers, and their relationships.

Enterprise Description: Automobile Company

You are modeling an automobile company such as General Motors, Ford, Toyota, or Volkswagen. In our hypothetical company, it has been decided to redesign a major part of the database that underlies company operations. Below are some suggested entities and relationships to consider:

- **Vehicles:** Each vehicle has a Vehicle Identification Number (VIN). Additional attributes may include the manufacturing date, assembly plant, and current status (e.g., in production, in inventory, sold).
- **Brands:** Each company may have several brands (for example, GM has Chevrolet, Buick, Cadillac, GMC, etc., and Volkswagen has Volkswagen, Audi, Lamborghini, Bentley, etc.). Each brand has a unique identifier and name.
- **Models:** Each brand offers several models (for example, Buick's models include the Enclave and LaCrosse). Each model may come in a variety of body styles (e.g., 4-door sedan, wagon, convertible, SUV). You may record the model name, year, and base price.
- **Options:** Vehicles may have various options such as color, engine type, and transmission type. You may model options as attributes or as separate entities depending on your design choices.
- **Suppliers:** External suppliers supply certain parts for certain models. A supplier may have multiple plants. You might record each supplier's name, contact information, and unique ID.
- **Manufacturing Plants:** The company owns manufacturing plants. Some plants produce specific parts for certain models, while others perform final assembly of vehicles. Each plant has a plant ID, name, and location.
- **Dealers:** Dealers sell vehicles to customers. Each dealer has a dealer ID, name, address, and may maintain an inventory of vehicles.

- **Customers:** Customers purchase vehicles through dealers. Customer attributes include name, address, phone, gender, and annual income.
- **Sales Transactions:** For the conceptual design, you may include a sales entity that records the date, dealer, vehicle sold, customer, payment method, and sale price.

Sample Query Requirements (Natural Language)

Although you are not required to implement these queries in SQL, your E-R diagram should make it **possible** to answer the following (or similar) questions. These are stated in natural language:

1. **Sales Trends:** *"Show sales trends for various brands over the past 3 years, by year, month, and week. Then break these data out by gender of the buyer and then by income range."*
2. **Defective Part Tracking:** *"Suppose that it is found that transmissions made by supplier Getrag between two given dates are defective. Find the VIN of each car containing such a transmission and the customer to which it was sold. If your design allows, suppose the defective transmissions all come from only one of Getrag's plants."*
3. **Top Brands by Revenue:** *"Find the top 2 brands by dollar-amount sold in the past year."*
4. **Top Brands by Unit Sales:** *"Find the top 2 brands by unit sales in the past year."*
5. **Seasonal Sales Patterns:** *"In what month(s) do convertibles sell best?"*
6. **Dealer Inventory Efficiency:** *"Find those dealers who keep a vehicle in inventory for the longest average time."*
7. **Supplier Coverage:** *"Which supplier provides parts for the most number of distinct models across all brands?"*

Design your E-R diagram so that these (and similar) queries can be answered intuitively. This may influence your decisions regarding entities, attributes, and relationships.

Important Considerations for E-R Design

When designing your E-R diagram, consider the following:

- **Entities vs. Attributes:** Determine which concepts should be modeled as entities (e.g., *Vehicle*, *Brand*, *Model*, *Supplier*) and which should be attributes (e.g., *color*, *engine type*).
- **Primary Keys and Cardinalities:** Identify primary key attributes (such as *VIN* or *supplier_ID*) for each entity. Define relationship cardinalities (1:1, 1:N, M:N) based on business rules (e.g., "A brand offers many models; each model belongs to exactly one brand").
- **M:N vs. 1:N Relationships:**
 - **1:N Example:** *One brand offers many models, but each model belongs to exactly one brand.*
 - **M:N Example:** *A supplier can supply parts for many models, and each model may use parts from many suppliers.*
- **Optional vs. Mandatory Participation:**
 - If some models may not yet have any vehicles manufactured, model that relationship as optional on the vehicle side.
 - If every vehicle must be associated with exactly one model, reflect this constraint in the diagram.

Deliverables

By the due date, submit the following:

1. **Final E-R Diagram**

Create a clear, well-labeled E-R diagram. Be sure to include:

- Entities, attributes, primary keys, foreign keys
- Relationships, mapping cardinality, and participation constraints

2. **Short Description / Business Rules**

Write a brief (one to two paragraphs) summary of the key assumptions and rules that guided your design. For example, explain how you modeled the brand-to-model hierarchy, supplier-to-part relationships, or vehicle assembly tracking.

3. **Coverage of Sample Queries**

In 3–5 sentences, explain how your diagram supports the sample queries provided (e.g., "Since each vehicle is linked to its model, brand, and the parts supplied by specific suppliers, we can trace defective parts back to the supplier and identify affected customers.").

Report Specification

The submitted report must include the following components:

- **Entity and Relationship Justification:** Provide a detailed explanation of why each entity and relationship was defined as such in your E-R diagram. Discuss the design choices made and how they reflect the underlying business logic.
- **Mapping Cardinality Explanation:** Explain the cardinality assigned to each relationship (e.g., 1:1, 1:N, M:N). Justify your decisions based on real-world assumptions and constraints within an automobile company.
- **Query Support Description:** Describe how your E-R model enables processing of the sample queries listed in the project specification. Reference specific parts of your diagram to explain how each query could be answered conceptually.
- **Format:** There is no fixed format for the report. You are free to structure and format it as you see fit, as long as all required content is clearly addressed.

ERD Specification

The E-R diagram (ERD) you submit must adhere to the following guidelines:

- **Tool:** The ERD should be created using Lucidchart (<https://www.lucidchart.com>).
- **Single View:** Ensure that the entire ERD is visible on a single screen without the need for scrolling or zooming. Organize your layout clearly and compactly.
- **File Format:** Submit the ERD as a .png file. Other formats (e.g., PDF, JPG) will not be accepted.
- **Clarity:** All notations in the ERD are the same as those in the textbook. See slide 6.43.

Grading

Your submission will be evaluated based on the following criteria:

- **Accuracy and Completeness (40%)**
 - Does the E-R diagram include all major entities (vehicles, brands, models, suppliers, etc.)?
 - Are the relationships and cardinalities consistent with the business domain?
- **Ability to Address Queries (40%)**
 - Can the E-R model support the sample queries in principle?
 - Are appropriate entities and relationships present to answer those queries?
- **Clarity and Presentation (20%)**
 - Overall readability and structure of the E-R diagram
 - Use of consistent and accepted notation (e.g., crow's foot, Chen)
 - Clear and coherent explanation of the business rules

Note: Actual implementation (e.g., SQL scripts or database setup) is not required. This project focuses **solely** on conceptual design.